# Lesson 3.1 Half-fraction of the $2^k$ Design

## Introduction

As the number of factors in a $2^k$ factorial design increases, the number of runs required for a complete replicate of the design rapidly outgrows the resources of most experimenters. For example, a complete replicate of the $2^6$ design requires 64 runs. In this design only 6 of the 63 degrees of freedom correspond to main effects, and only 15 degrees of freedom correspond to two-factor interactions. There are only 21 degrees of freedom associated with effects that are likely to be of major interest. The remaining 42 degrees of freedom are associated with three-factor and higher interactions.

If the experimenter can reasonably assume that certain high-order interactions are negligible, information on the main effects and low-order interactions may be obtained by running only a fraction of the complete factorial experiment. These **fractional factorial** designs are among the most widely used types of designs for product and process design, process improvement, and industrial/business experimentation.

A major use of fractional factorials is in screening experiments—experiments in which many factors are considered and the objective is to identify those factors (if any) that have large effects. Screening experiments are usually performed in the early stages of a project when many of the factors initially considered likely have little or no effect on the response. The factors identified as important are then investigated more thoroughly in subsequent experiments.

The successful use of fractional factorial designs is based on three key ideas:

1. The **sparsity of effects** principle. When there are several variables, the system or process is likely to be driven primarily by some of the main effects and low-order interactions.

2. The **projection property**. Fractional factorial designs can be projected into stronger (larger) designs in the subset of significant factors.

3. **Sequential experimentation**. It is possible to combine the runs of two (or more) fractional factorials to construct sequentially a larger design to estimate the factor effects and interactions of interest.

## The One-Half Fraction of the $2^k$ Design

Consider a situation in which three factors, each at two levels, are of interest, but the experimenters cannot afford to run all $2^3 = 8$ treatment combinations. They can, however, afford four runs. This suggests a one-half fraction of a $2^3$ design. Because the design contains $2^{3-1} = 4$ treatment combinations, a one-half fraction of the $2^3$ design is often called a $2^{3-1}$ design.

The table of plus and minus signs for the $2^3$ design is shown below. Suppose we select the four treatment combinations $a$, $b$, $c$, and $abc$ as our one-half fraction. Notice that the $2^{3-1}$ design is formed by selecting only those treatment combinations that have a plus in the $ABC$ column. Thus, $ABC$ is called the generator of this particular fraction. Usually we will refer to a generator such as ABC as a **word**.

**Plus and Minus Signs for the $2^3$ Factorial Design**

| Treatment Combination | Factorial Effect | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | $I$ | $A$ | $B$ | $C$ | $AB$ | $AC$ | $BC$ | $ABC$ |
| $a$ | + | + | − | − | − | − | + | + |
| $b$ | + | − | + | − | − | + | − | + |
| $c$ | + | − | − | + | + | − | − | + |
| $abc$ | + | + | + | + | + | + | + | + |
| $ab$ | + | + | + | − | + | − | − | − |
| $ac$ | + | + | − | + | − | + | − | − |
| $bc$ | + | − | + | + | − | − | + | − |
| (1) | + | − | − | − | + | + | + | − |

Furthermore, the identity column $I$ is also always plus, so we call

$$I = ABC$$

the defining relation for our design. In general, the defining relation for a fractional factorial will always be the set of all columns that are equal to the identity column $I$.

The treatment combinations in the $2^{3-1}$ design yield three degrees of freedom that we may use to estimate the main effects. Note that the linear combinations of the observations used to estimate the main effects of $A$, $B$, and $C$ are

$$[A] = \frac{1}{2}(a - b - c + abc)$$
$$[B] = \frac{1}{2}(-a + b - c + abc)$$
$$[C] = \frac{1}{2}(-a - b + c + abc)$$

Where the notation $[A]$, $[B]$, and $[C]$ is used to indicate the linear combinations associated with the main effects. It is also easy to verify that the linear combinations of the observations

used to estimate the two-factor interactions are

$$[AB] = \frac{1}{2}(-a - b + c + abc)$$
$$[AC] = \frac{1}{2}(-a + b - c + abc)$$
$$[BC] = \frac{1}{2}(a - b - c + abc)$$

Thus, $[A] = [BC]$, $[B] = [AC]$, and $[C] = [AB]$; consequently, it is impossible to differentiate between $A$ and $BC$, $B$ and $AC$, and $C$ and $AB$. In fact, when we estimate $A$, $B$, and $C$ we are really estimating $A + BC$, $B + AC$, and $C + AB$. Two or more effects that have this property are called **aliases**.

In our example, $A$ and $BC$ are aliases, $B$ and $AC$ are aliases, and $C$ and $AB$ are aliases. We indicate this by the notation $[A] \to A + BC$, $[B] \to B + AC$, and $[C] \to C + AB$.

The alias structure for this design may be easily determined by using the defining relation $I = ABC$. Multiplying any column (or effect) by the defining relation yields the aliases for that column (or effect). In our example, this yields as the alias of $A$

$$A \cdot I = A \cdot ABC = A^2BC = BC$$

because the square of any column is just the identity $I$. The alias of $B$ and $C$ are obtained similarly.

This one-half fraction, with $I = +ABC$, is usually called the **principal fraction**.

Now suppose that we had chosen the other one-half fraction, that is, the treatment combinations associated with minus in the ABC column. This **alternate**, or **complementary**, one-half fraction (consisting of the runs (1), $ab$, $ac$, and $bc$)

The defining relation for this design is

$$I = -ABC$$

The linear combination of the observations, say $[A]'$ , $[B]'$ , and $[C]'$ , from the alternate fraction gives us

$$[A]' \to A - BC$$
$$[B]' \to B - AC$$
$$[C]' \to C - AB$$

Thus, when we estimate $A$, $B$, and $C$ with this particular fraction, we are really estimating $A - BC$, $B - AC$, and $C - AB$.

In practice, it does not matter which fraction is actually used. Both fractions belong to the same family; that is, the two one-half fractions form a complete $2^3$ design.

Suppose that after running one of the one-half fractions of the 23 design, the other fraction was also run. Thus, all eight runs associated with the full 23 are now available. We may now obtain de-aliased estimates of all the effects by analyzing the eight runs as a full $2^3$ design in two blocks of four runs each. This could also be done by adding and subtracting the linear combination of effects from the two individual fractions. For example, consider $[A] \to A + BC$ and $[A]' \to A - BC$. This implies that

$$\frac{1}{2}([A] + [A]') = 2(A + BC + A - BC) \to A$$

and that

$$\frac{1}{2}([A] - [A]') = 2(A + BC - A + BC) \to BC$$

Thus, for all three pairs of linear combinations, we would obtain the following:

| $i$ | From $\frac{1}{2}([i] + [i]')$ | From $\frac{1}{2}([i] - [i]')$ |
|---|---|---|
| $A$ | $A$ | $BC$ |
| $B$ | $B$ | $AC$ |
| $C$ | $C$ | $AB$ |

Furthermore, by assembling the full $2^3$ in this fashion with $I = +ABC$ in the first group of runs and $I = -ABC$ in the second, the $2^3$ confounds $ABC$ with blocks.

## Design Resolution

The preceding $2^{3-1}$ design is called a resolution III design. In such a design, main effects are aliased with two-factor interactions. A design is of resolution $R$ if no $p$-factor effect is aliased with another effect containing less than $R - p$ factors. We usually employ a Roman numeral subscript to denote design resolution; thus, the one-half fraction of the $2^3$ design with the defining relation $I = +ABC$ (or $I = -ABC$) is a $2^{3-1}_{III}$ design.

Designs of resolution III, IV, and V are particularly important. The definitions of these designs and an example of each follow:

1. **Resolution III** designs. These are designs in which no main effects are aliased with any other main effect, but main effects are aliased with two-factor interactions and some two-factor interactions may be aliased with each other.

2. **Resolution IV** designs. These are designs in which no main effect is aliased with any other main effect or with any two-factor interaction, but two-factor interactions are aliased with each other. An example of this design is a $2^{4-1}$ design with $I = ABCD$.

3. **Resolution V** designs. These are designs in which no main effect or two-factor interaction is aliased with any other main effect or two-factor interaction, but two factor interactions are aliased with three-factor interactions. A $2^{5-1}$ design with $I = ABCDE$ is a resolution V design.

In general, the resolution of a two-level fractional factorial design is equal to the number of letters in the shortest word in the defining relation. Consequently, we could call the preceding design types three-, four-, and five-letter designs, respectively. We usually like to employ fractional designs that have the highest possible resolution consistent with the degree of fractionation required. The higher the resolution, the less restrictive the assumptions that are required regarding which interactions are negligible to obtain a unique interpretation of the results.

## Construction and Analysis of the One-Half Fraction

A one-half fraction of the $2^k$ design of the highest resolution may be constructed by writing down a basic design consisting of the runs for a full $2^{k-1}$ factorial and then adding the $k^{th}$ factor by identifying its plus and minus levels with the plus and minus signs of the highest order interaction $ABC \cdots (K-1)$. Therefore, the $2^{3-1}_{III}$ fractional factorial is obtained by writing down the full $2^2$ factorial as the basic design and then equating factor $C$ to the $+AB$ interaction. The alternate fraction would be obtained by equating factor $C$ to the $-AB$ interaction. This approach is illustrated in the table below.

**The Two One-Half Fractions of the $2^3$ Design**

| | Full $2^2$ Factorial (Basic Design) | | $2^{3-1}_{III}$, $I = ABC$ | | | $2^{3-1}_{III}$, $I = -ABC$ | | |
|---|---|---|---|---|---|---|---|---|
| **Run** | ***A*** | ***B*** | ***A*** | ***B*** | ***C = AB*** | ***A*** | ***B*** | ***C = − AB*** |
| 1 | − | − | − | − | + | − | − | − |
| 2 | + | − | + | − | − | + | − | + |
| 3 | − | + | − | + | − | − | + | + |
| 4 | + | + | + | + | + | + | + | − |

Notice that the basic design always has the right number of runs (rows), but it is missing one column. The generator $I = ABC \cdots K$ is then solved for the missing column ($K$) so that $K = ABC \cdots (K-1)$ defines the product of plus and minus signs to use in each row to produce the levels for the $k^{th}$ factor.

Note that any interaction effect could be used to generate the column for the $k^{th}$ factor. However, using any effect other than $ABC \cdots (K-1)$ will not produce a design of the highest possible resolution.

Another way to view the construction of a one-half fraction is to partition the runs into two blocks with the highest order interaction $ABC \cdots K$ confounded. Each block is a $2^{k-1}$fractional factorial design of the highest resolution.

**Example**

Consider the filtration example in Lesson 2.1 where a factorial experiment was carried out in the pilot plant to study the factors thought to influence the filtration rate of this product. The four factors are temperature (A), pressure (B), concentration of formaldehyde (C), and stirring rate (D). The data represents a single replicate of the $2^4$.

In that example, we found that the main effects A, C, and D and the interactions AC and AD were different from zero. We will now return to this experiment and simulate what would have happened if a half-fraction of the $2^4$ design had been run instead of the full factorial.

We will use the $2^{4-1}$ design with $I = ABCD$, because this choice of generator will result in a design of the highest possible resolution (IV). To construct the design, we first write down the basic design, which is a $2^3$ design, as shown in the first three columns of the table below.

**The $2^{4-1}_{IV}$ Design with the Defining Relation $I$ = ABCD**

| | Basic Design | | | | | |
|---|---|---|---|---|---|---|
| **Run** | ***A*** | ***B*** | ***C*** | ***D = ABC*** | **Treatment Combination** | **Filtration Rate** |
| 1 | − | − | − | − | (1) | 45 |
| 2 | + | − | − | + | *ad* | 100 |
| 3 | − | + | − | + | *bd* | 45 |
| 4 | + | + | − | − | *ab* | 65 |
| 5 | − | − | + | + | *cd* | 75 |
| 6 | + | − | + | − | *ac* | 60 |
| 7 | − | + | + | − | *bc* | 80 |
| 8 | + | + | + | + | *abcd* | 96 |

This basic design has the necessary number of runs (eight) but only three columns (factors). To find the fourth factor levels, solve $I = ABCD$ for $D$, or $D = ABC$. Thus, the level of $D$ in each run is the product of the plus and minus signs in columns A, B, and C. Because the generator $ABCD$ is positive, this design is the principal fraction.

Using the defining relation, we note that each main effect is aliased with a three-factor interaction; that is, $A = A^2BCD = BCD$, $B = AB^2CD = ACD$, $C = ABC^2D = ABD$, and $D = ABCD^2 = ABC$.

Furthermore, every two-factor interaction is aliased with another two-factor interaction. These alias relationships are $AB = CD$, $AC = BD$, and $BC = AD$. The four main effects plus the three two-factor interaction alias pairs account for the seven degrees of freedom for the design.

At this point, we would normally randomize the eight runs and perform the experiment. Because we have already run the full $2^4$ design, we will simply select the eight observed filtration rates from Lesson 2.1 that correspond to the runs in the design. These observations are shown in the last column of the previous table.

The effects can be calculated as follows:

$$\frac{contrast}{r(2^{k-p-1})}$$

For example,

$$[A] = \frac{1}{4}(-45 + 100 - 45 + 65 - 75 + 60 - 80 + 96) = 19 \rightarrow A + BCD$$

and

$$[AB] = \frac{1}{4}(45 - 100 - 45 + 65 + 75 - 60 - 80 + 96) = -1 \rightarrow AB + CD$$

The complete list of effect estimates are shown below.

| Estimate | Alias Structure |
|---|---|
| **[A] = 19.00** | $[A] \rightarrow A + BCD$ |
| [B] = 1.50 | $[B] \rightarrow B + ACD$ |
| **[C] = 14.00** | $[C] \rightarrow C + ABD$ |
| **[D] = 16.50** | $[D] \rightarrow D + ABC$ |
| [AB] = −1.00 | $[AB] \rightarrow AB + CD$ |
| **[AC] = −18.50** | $[AC] \rightarrow AC + BD$ |
| **[AD] = 19.00** | $[AD] \rightarrow AD + BC$ |

[a] Significant effects are shown in boldface type.

From inspection of the information in the above table, it is not unreasonable to conclude that the main effects $A$, $C$, and $D$ are large. The $AB+CD$ alias chain has a small estimate, so the simplest interpretation is that both the $AB$ and $CD$ interactions are negligible (otherwise, both $AB$ and $CD$ are large, but they have nearly identical magnitudes and opposite signs—this is fairly unlikely). Furthermore, if $A$, $C$, and $D$ are the important main effects, then it is logical to conclude that the two interaction alias chains $AC + BD$ and $AD + BC$ have large effects because the $AC$ and $AD$ interactions are also significant. In other words, if $A$, $C$, and $D$ are significant then the significant interactions are most likely $AC$ and $AD$. This is an application of **Ockham's razor** (after William of Ockham), a scientific principle that when one is confronted with several different possible interpretations of a phenomena, the simplest interpretation is usually the correct one.

Another way to view this interpretation is from the standpoint of effect *heredity*. Suppose that $AB$ is significant and that both main effects $A$ and $B$ are significant. This is called **strong heredity**, and it is the usual situation (if an interaction is significant and only one of the main effects is significant this is called **weak heredity**; and this is relatively less common). So in this example, with $A$ significant and $B$ not significant this support the assumption that $AB$ is not significant.

The following code chunks generates the design and calculates the effect estimates.

```
library(tidyverse)
library(FrF2)

#generates the design matrix
set.seed(1234)
design <- as.data.frame(FrF2(nruns = 8,
               nfactors = 4,
               generators = "ABC"))
design
```

```
##    A  B  C  D
## 1  1  1 -1 -1
## 2  1 -1 -1  1
## 3  1 -1  1 -1
## 4 -1 -1  1  1
## 5  1  1  1  1
## 6 -1  1 -1  1
## 7 -1 -1 -1 -1
## 8 -1  1  1 -1
```

```
rate <- c(65, 100, 60, 75, 96, 45, 45, 80)

df <- design %>%
  mutate(Y=rate)
df
```

```
##    A  B  C  D   Y
## 1  1  1 -1 -1  65
## 2  1 -1 -1  1 100
## 3  1 -1  1 -1  60
## 4 -1 -1  1  1  75
## 5  1  1  1  1  96
## 6 -1  1 -1  1  45
## 7 -1 -1 -1 -1  45
## 8 -1  1  1 -1  80
```

```
model <- lm(Y ~ A*B*C, data = df)

summary(model)
```

```
##
```

```
## Call:
## lm.default(formula = Y ~ A * B * C, data = df)
##
## Residuals:
## ALL 8 residuals are 0: no residual degrees of freedom!
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    70.75        NaN     NaN      NaN
## A1              9.50        NaN     NaN      NaN
## B1              0.75        NaN     NaN      NaN
## C1              7.00        NaN     NaN      NaN
## A1:B1          -0.50        NaN     NaN      NaN
## A1:C1          -9.25        NaN     NaN      NaN
## B1:C1           9.50        NaN     NaN      NaN
## A1:B1:C1        8.25        NaN     NaN      NaN
##
## Residual standard error: NaN on 0 degrees of freedom
## Multiple R-squared:      1,  Adjusted R-squared:    NaN
## F-statistic:   NaN on 7 and 0 DF,  p-value: NA
```

**Example**

Suppose we run the other half, that is, with $I = -ABCD$. Manually generate the design and calculate the effect estimates.

[LEFT as a CLASSROOM EXERCISE!]